Universal Format

Input File Specification

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1. Introduction

The Universal Format is an input format designed to describe the geometry and material properties of both 2D cross-sections and 3D layouts for electrostatic field simulations. It was developed to serve as a common interface across multiple field solvers, both open-source and industrial.

2. Format Specification

The format defines four building blocks: a ground plane, one or more dielectric layers, wire geometry, and the simulation window. Each block is expressed as a single directive line, with separate syntax for 2D and 3D input files.

2.a. Ground Plane

Defines an infinite ground plane placed at a specified vertical height. Typically used to model the substrate or a reference plane for electrostatic potential.

2D Input File

```
GROUND_PLANE <index> <name> HEIGHT <ylow> <yhigh> THICKNESS <thickness>
```

3D Input File

```
GROUND_PLANE <index> <name> HEIGHT <zlow> <zhigh> THICKNESS <thickness>
```

Field	Description
index	Integer identifier for the ground plane.
name	Descriptive name for the ground plane (e.g. layer/net name).
ylow / yhigh (2D) zlow / zhigh (3D)	Vertical extent of the ground plane along the cross-section axis.
thickness	Physical thickness of the ground plane.

2.b. Dielectric Layers

Specifies vertically stacked dielectric materials. These regions influence the field solution based on their relative permittivity. In both 2D and 3D formats, dielectric regions extend along the x-axis, and their horizontal span is determined by the coordinates of the simulation bounding box defined in Section 2.d (Simulation Window).

2D Input File

```
DIELECTRIC <name> HEIGHT <ylow> <yhigh> EPSILON <relative_permittivity>
```

3D Input File

```
DIELECTRIC <name> HEIGHT <zlow> <zhigh> EPSILON <relative_permittivity>
```

Field	Description
name	Descriptive name of the dielectric layer (e.g. process stack label).
ylow / yhigh (2D) zlow / zhigh (3D)	Vertical extent of the dielectric layer.
relative_permittivity	Relative permittivity (epsilon) of the dielectric material.

2.c. Wire Geometry

Describes a rectangular conductive wire using its four corners in the vertical plane (2D) or vertical cross-section (3D). Each wire is assigned a voltage, though this parameter is currently unused in FasterCap. It is preserved for compatibility with the Raphael format and for potential future use.

2D Input File

```
WIRE <id> <name> LL <x> <y> LR <x> <y> UR <x> <y> UL <x> <y> VOLTAGE <v>
```

3D Input File

```
WIRE <id> <name> LL <x> <z> LR <x> <z> UR <x> <z> UL <x> <z> LENGTH <y> VOLTAGE <v>
```

Field	Description
id	Integer identifier for the wire.
name	Descriptive name for the wire (e.g. layer and instance label).
LL / LR / UR / UL	Coordinates of the lower-left, lower-right, upper-right, and upper-left corners of the wire's rectangular cross-section.
LENGTH (3D only)	Extent of the wire along the y-axis (out-of-plane direction).
VOLTAGE	Assigned voltage. Currently unused by FasterCap; retained for Raphael-format compatibility and future use.

2.d. Simulation Window

Defines the simulation region's bounding box and an optional extended window. All wires and dielectric structures must lie within this window.

2D Input File

```
WINDOW_BBOX LL <x1> <y1> UR <x2> <y2> SIM_WIN_EXT LL <x1> <y1> UR <x2> <y2>
```

3D Input File

```
WINDOW_BBOX LL <x1> <z1> UR <x2> <z2> LENGTH <max_y> SIM_WIN_EXT LL <x1> <z1> UR <x2>
<z2> LENGTH <min_y> <max_y>
```

Field	Description
WINDOW_BBOX	Lower-left (LL) and upper-right (UR) corners of the primary simulation bounding box. The 3D form adds a LENGTH extent along y.
SIM_WIN_EXT	Optional extended simulation window, using the same corner convention. The 3D form specifies both a minimum and maximum y extent.

3. Example 2D Input File

```
PATTERN 2D-Universal/DiagUnder6/M1duM5/D0/W0.7/S0.28/wires


# Ground Plane(s)
GROUND_PLANE 0 M0__GND HEIGHT -1.000 0.000 THICKNESS 1.000


# Process Stack -- Dielectric Heights
DIELECTRIC p1_1 HEIGHT 0.000 0.250 EPSILON 4.100
DIELECTRIC m1_1 HEIGHT 0.250 0.300 EPSILON 7.000
DIELECTRIC m1_2 HEIGHT 0.300 0.650 EPSILON 4.300
DIELECTRIC m1_3 HEIGHT 0.650 0.700 EPSILON 4.300
DIELECTRIC m1_4 HEIGHT 0.700 0.750 EPSILON 4.900


# Wires
WIRE 1 M1_w1 LL -1.750 0.650 LR -1.610 0.650 UR -1.610 0.900 UL -1.750 0.900 VOLTAGE 0
WIRE 2 M1_w2 LL -1.470 0.650 LR -1.330 0.650 UR -1.330 0.900 UL -1.470 0.900 VOLTAGE 0


# Simulation Window
WINDOW_BBOX LL -50.000 0.000 UR 50.000 19.056
SIM_WIN_EXT LL 0.000 0.000 UR 0.000 0.000 LENGTH 0.000 0.000
```

4. Example 3D Input File

```
PATTERN UNIVERSAL_3D_PATTERNS_MODIFIED/Over6/M1oM0/D0/W0.14/S0.7/wires


# Ground Plane(s)
GROUND_PLANE 0 M0__w0 HEIGHT -1.000 0.000 THICKNESS 1.000


# Process Stack -- Dielectric Heights
DIELECTRIC p1_1 HEIGHT 0.000 0.250 EPSILON 4.100
DIELECTRIC m1_1 HEIGHT 0.250 0.300 EPSILON 7.000
DIELECTRIC m1_2 HEIGHT 0.300 0.650 EPSILON 4.300
DIELECTRIC m1_3 HEIGHT 0.650 0.700 EPSILON 4.300
DIELECTRIC m1_4 HEIGHT 0.700 0.750 EPSILON 4.900


# Wires
WIRE 5 M1_w5 LL 0.770 0.650 LR 0.910 0.650 UR 0.910 0.900 UL 0.770 0.900 LENGTH 2.000
VOLTAGE 0
WIRE 6 M1_w6 LL 1.050 0.650 LR 1.190 0.650 UR 1.190 0.900 UL 1.050 0.900 LENGTH 2.000
VOLTAGE 0


# Simulation Window
WINDOW_BBOX LL -10.000 0.000 UR 10.000 19.056 LENGTH 20.000
SIM_WIN_EXT LL 0.000 0.000 UR 0.000 0.000 LENGTH 0.000 0.000
```